

● EASY

● ALGEBRA

Linear functions

01

10 questions · ~15 min

Q1

ALGEBRA

If f is the function defined by $f(x) = \frac{2x-1}{3}$, what is the value of $f(5)$?

A. $\frac{4}{3}$

B. $\frac{7}{3}$

C. 3

D. 9

Q2

GRID-IN ALGEBRA

The function g is defined by $gx = 6x$. For what value of x is $g(x) = 54$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN

ALGEBRA

The function f is defined by $fx = 4x$. For what value of x does $fx = 8$?

STUDENT-PRODUCED RESPONSE

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.	/	.	/	.
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0	0	0	0
---	---	---	---

1	1	1	1
---	---	---	---

2	2	2	2
---	---	---	---

3	3	3	3
---	---	---	---

4	4	4	4
---	---	---	---

5	5	5	5
---	---	---	---

6	6	6	6
---	---	---	---

7	7	7	7
---	---	---	---

8	8	8	8
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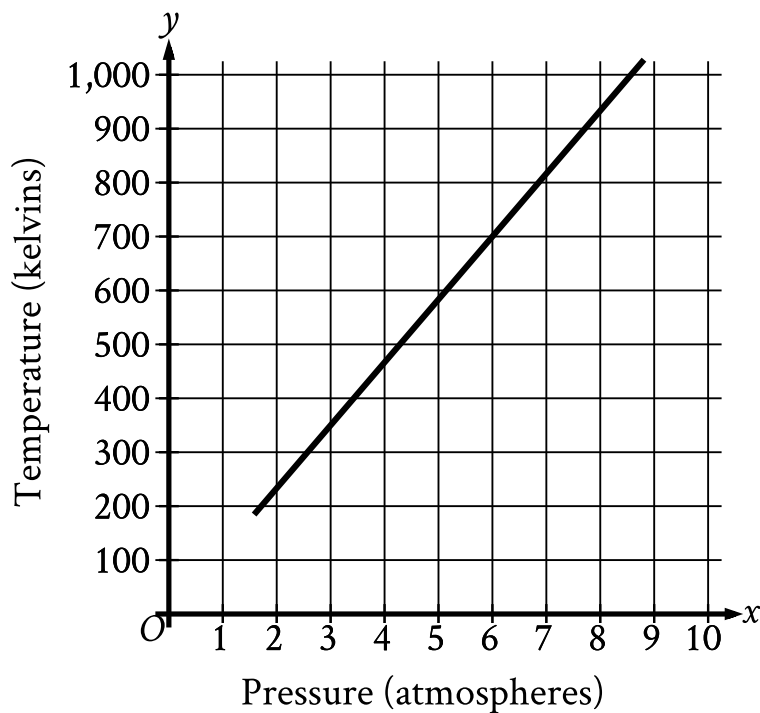
9	9	9	9
---	---	---	---

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Gabriella deposits \$ 35 in a savings account at the end of each week. At the beginning of the 1st week of a year there was \$ 600 in that savings account. How much money, in dollars, will be in the account at the end of the 4th week of that year?

- A. 460
- B. 635
- C. 639
- D. 740

Oxygen gas is placed inside a tank with a constant volume. The graph shows the estimated temperature y , in kelvins, of the oxygen gas when its pressure is x atmospheres.



- The line slants sharply up from left to right.
- The line begins at the approximate point (1.6 comma 190).
- The line passes through the following approximate points:
 - (3 comma 350)
 - (6 comma 700)
 - (8 comma 933)

What is the estimated temperature, in kelvins, of the oxygen gas when its pressure is 6 atmospheres?

- A. 6
- B. 60

C. 700

D. 760

Q6

ALGEBRA

The number y is 84 less than the number x . Which equation represents the relationship between x and y ?

A. $y = x + 84$

B. $y = \frac{1}{84}x$

C. $y = 84x$

D. $y = x - 84$

x	$f(x)$
1	5
3	13
5	21

Some values of the linear function f are shown in the table above. Which of the following defines f ?

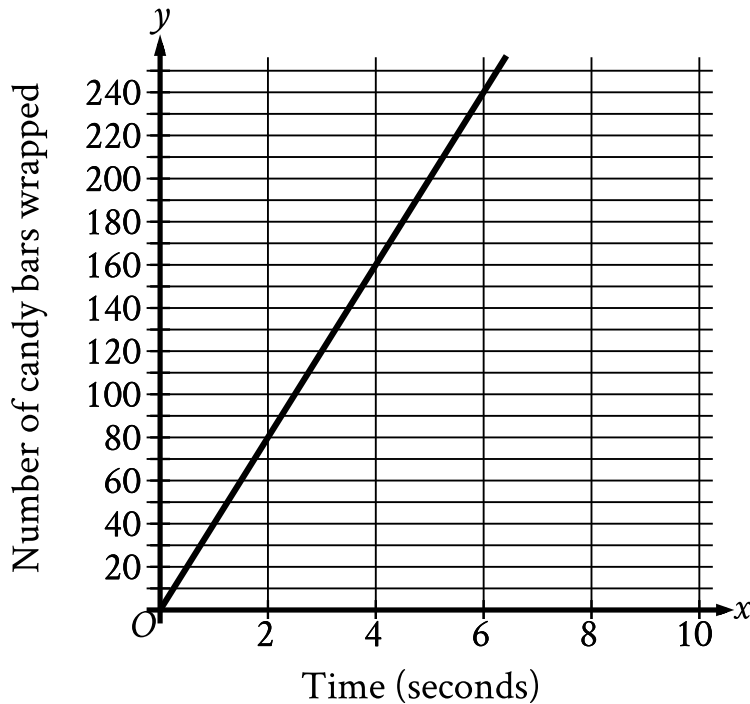
A. $f(x) = 2x + 3$

B. $f(x) = 3x + 2$

C. $f(x) = 4x + 1$

D. $f(x) = 5x$

The graph shown models the number of candy bars a certain machine wraps with a label in x seconds.



- The line slants sharply up from left to right.
- The line passes through the following points:
 - (0 comma 0)
 - (2 comma 80)
 - (4 comma 160)
 - (6 comma 240)

According to the graph, what is the estimated number of candy bars the machine wraps with a label per second?

- A. 2
- B. 40

C. 78

D. 80

Q9

ALGEBRA

The function f is defined by the equation $fx = 100x + 2$. What is the value of fx when $x = 9$?

A. 111

B. 118

C. 900

D. 902

Q10

ALGEBRA

$$f(x) = 4x + b$$

For the linear function f , b is a constant and $f(7) = 28$. What is the value of b ?

A. 0

B. 1

C. 4

D. 7